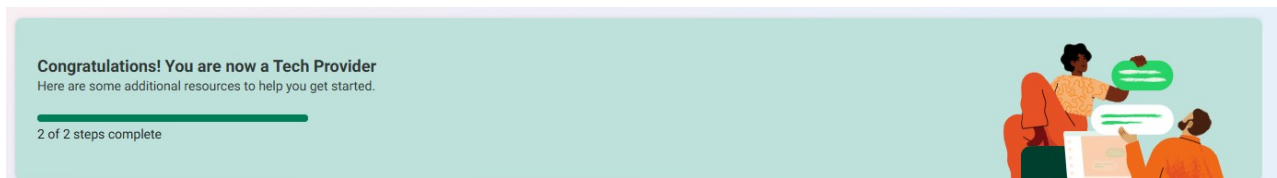


Slotly WhatsApp Automation Platform

Detailed build plan for a self-service, multi-tenant WhatsApp SaaS

Confirmed starting position

Meta dashboard confirmation supplied by the project owner: “Congratulations! You are now a Tech Provider” and “2 of 2 steps complete.” SlotlyDev can therefore proceed with the Tech Provider / Embedded Signup implementation path. The production implementation must still be tested end-to-end with unrelated external businesses.



Meta Tech Provider confirmation - user-provided screenshot, 19 August 2026

Primary objective

Take an unrelated business from “I use WhatsApp for my business” to “Slotly can securely send and receive messages for my business” without SlotlyDev manually configuring that customer’s Meta account.

Plan at a glance

- 1. Executive decision and product principle
 - 2. Confirmed Meta position and assumptions
 - 3. MVP scope: what is in and what is deliberately out
 - 4. Customer prerequisites and onboarding promise
 - 5. Technical architecture and tenant model
 - 6. Phase 0: production Meta readiness
 - 7. Phase 1: account + Embedded Signup
 - 8. Phase 2: automatic post-signup provisioning
 - 9. Phase 3: webhooks, send/receive and tenant routing
 - 10. Phase 4: external-business and Business App onboarding tests
 - 11. Phase 5: pilot-ready connection dashboard
 - 12. Phase 6: first sellable automation service
 - 13. Data model, security and observability
 - 14. Test matrix and failure handling
 - 15. Pilot, metrics and go/no-go gates
 - 16. Delivery sequence, estimated timeline and backlog
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Executive decision

Build the onboarding and connection layer first. Do not begin with AI, a CRM, broadcasts or a visual automation builder. The highest-risk assumption is not whether a bot can reply; it is whether a normal external business can connect through Slotly with minimal assistance and whether Slotly can automatically complete the technical provisioning afterward.

The first product is not a chatbot

Internally, treat the initial release as “Slotly WhatsApp Onboarding MVP”. Its job is to connect, provision, receive, route and send. Only after this works reliably should the first customer-facing automation service be added.

Product principle and definition of “self-service”

The platform should hide the WhatsApp Business Platform infrastructure from the business owner. The customer may still need to complete Meta-controlled actions such as authentication, business selection, authorization, number verification and other eligibility steps. “Self-service” therefore means Slotly does not manually configure the customer’s Meta Business account or WhatsApp assets; it does not mean the customer performs zero actions.

Owner	Definition
Customer responsibility	Log in to Meta, select or create eligible business assets, authorize Slotly, verify/control the phone number where Meta requires it, and complete any Meta-controlled eligibility/payment steps.
Slotly responsibility	Launch Embedded Signup, capture onboarding output, securely associate the assets with the correct tenant, perform supported post-signup provisioning, subscribe the correct WABA/webhook path, verify connection state, and provide clear recovery guidance.
Support responsibility	Exception path only. Support helps when Meta rejects, blocks or requires additional customer action. It must not be the normal onboarding path.

Non-negotiable product rule

No bespoke setup per customer

If every new business requires Rudy or another operator to open Meta Business Manager, copy IDs, create webhooks or configure credentials manually, the product has become an agency workflow rather than scalable SaaS.

Success statement for the first milestone

A business that is unrelated to SlotlyDev can create an account, click Connect WhatsApp, complete Meta’s onboarding flow, return to Slotly, and successfully send and receive a test message through Slotly without SlotlyDev manually touching the customer’s Meta account.

What is confirmed vs what still requires production proof

Item	Status	Meaning
Tech Provider status	CONFIRMED	Screenshot explicitly states the app/business is now a Tech Provider and 2 of 2 steps are complete.
WhatsApp app permissions	CONFIRMED FROM PROJECT CONTEXT	Advanced access for whatsapp_business_management and whatsapp_business_messaging has been approved in the existing project flow.
Embedded Signup path	SUPPORTED BY META	Meta documents Embedded Signup as the onboarding mechanism for Tech Providers and their business customers.
External customer onboarding	MUST BE PROVEN	Run a production-style test with an unrelated business. Internal app roles/test assets are not enough evidence.
Existing WhatsApp Business App number path	MUST BE PROVEN	Meta documents Business App onboarding; exact eligibility, current flow/version and account-specific restrictions must be tested.
Zero support for every customer	NOT ASSUMED	The product goal is high self-service completion, with a controlled exception path for Meta/account failures.

Meta source position

Meta's current documentation describes Embedded Signup as a flow that gathers information from business customers and automatically generates the WhatsApp assets needed by the platform. Meta also maintains separate documentation for the steps a Tech Provider performs after a customer completes Embedded Signup. This plan treats those official documents as the source of truth for implementation details.

What to build first

Scope	Contents
IN	Slotly account, business tenant, Connect WhatsApp button, Embedded Signup integration, onboarding-session state, capture of returned Meta identifiers, secure credential/token handling, post-signup provisioning, webhook subscription/routing, connection status, send test message, receive test message, failure/retry states.
OUT	AI assistant, shared team inbox, CRM, campaigns/broadcasts, visual flow builder, analytics dashboards, multiple channels, VoIP, advanced templates, payment collection, external integrations, complex role systems.

Why this boundary matters

- It isolates the single highest technical/product risk: external business onboarding.
- It prevents polished UI from hiding a broken provisioning path.
- It makes failures observable because there are fewer moving parts.
- It creates a reusable connection layer for every later product feature.
- It gives Slotly an asset that competitors may still handle through assisted setup.

Definition of done for the onboarding MVP

- An external business can complete the flow in production-like conditions.
- The correct business/WABA/phone identifiers are stored against the correct Slotly tenant.
- Slotly receives a webhook from that customer's WhatsApp account.
- Slotly can send a test reply from that customer's number.
- The customer sees a clear Connected state without technical IDs being exposed.
- Failed/cancelled onboarding can be safely retried without duplicate or corrupted tenant records.

What a business needs before onboarding

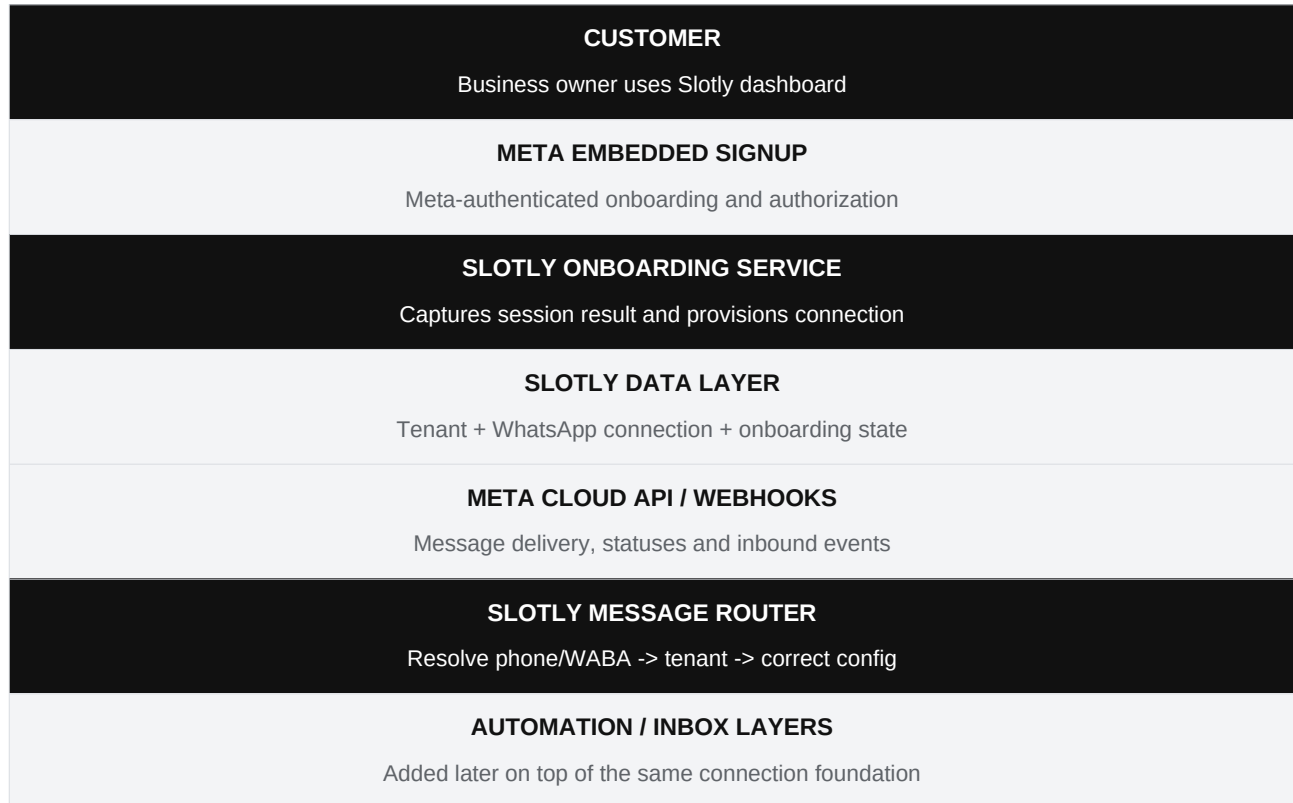
Prerequisite	Practical meaning
Meta login	A Facebook/Meta account the user can authenticate with during Embedded Signup.
Authority	The person completing signup must have sufficient access to the relevant business assets if selecting existing assets.
Business information	Business name and the details Meta requests during the onboarding flow.
Phone number control	A business number the customer is authorized to use and can verify when required.
Existing WhatsApp Business App	Potentially supported through Meta's Business App onboarding path; treat eligibility as a tested capability, not a blanket promise until production testing passes.
Meta billing / verification requirements	If Meta requests a payment method, business verification or other account action, the customer must complete it inside Meta. Slotly should explain the action but should not impersonate or bypass the customer.

Customer-facing wording should stay simple

"Have your Facebook login, business details and business WhatsApp number ready. Meta may ask you to confirm your business or number during connection." Avoid terms such as WABA, Graph API, webhook and access token in the customer UI.

Multi-tenant connection architecture

The platform should use one central application and messaging layer. Every incoming event is mapped to a Slotly tenant using Meta identifiers such as the phone number ID / WABA relationship. Do not deploy one application or webhook stack per customer.



Recommended application boundaries

- Web application: account, tenant setup, connection UI and status.
- Onboarding service: Embedded Signup session tracking, callback/event capture and post-signup provisioning.
- Webhook ingress: verify requests, persist raw event metadata, deduplicate, acknowledge quickly.
- Message router: map Meta identifiers to tenant and forward to the correct processing logic.
- Background worker/queue: retries, provisioning steps, outbound sends and non-blocking processing.
- Database: tenant ownership, WhatsApp connection records, onboarding sessions, webhook events, audit log.
- Secret store/encryption layer: long-lived credentials/tokens never exposed to the browser.

Production Meta readiness

Do this before building the polished onboarding UI. The objective is to make the Meta application configuration deterministic and document exactly which production configuration the code will rely on.

Step	Task	Acceptance condition
0.1	Confirm production app	Identify the exact Meta app that owns the approved Tech Provider / WhatsApp permissions. Avoid building against the wrong development/test app.
0.2	Record configuration IDs	Document the Embedded Signup configuration ID(s), app ID, webhook endpoint expectations and permitted production origins. Store values as environment configuration, not hard-coded UI values.
0.3	Review permission status	Confirm Advanced Access remains active for the WhatsApp permissions used by the flow.
0.4	Webhook verification	Set up a stable HTTPS webhook endpoint, verification token/logic and request-signature validation strategy.
0.5	Business App path decision	Decide whether Business App onboarding is in MVP or a controlled beta. Verify the current Meta flow/version before promising it publicly.
0.6	Create test matrix	Prepare one internal account and at least one unrelated external business/account for real validation.

Gate 0

Do not continue to product polish until the app configuration, Embedded Signup configuration and webhook endpoint are known and reproducible.

Slotly account + Embedded Signup

1. Create a Slotly user account and a Business/Tenant record.
2. Show a single primary action: Connect WhatsApp.
3. Create an onboarding session server-side before opening Meta. Give it a unique state/nonce and associate it with the current tenant.
4. Launch Meta Embedded Signup using the approved production configuration.
5. Capture the output/events supplied by the Embedded Signup implementation according to Meta's current documentation.
6. Validate that the returned onboarding data belongs to the expected session and logged-in tenant.
7. Persist only the identifiers/state required for the next provisioning steps. Do not expose credentials in browser logs or client-side state.
8. If the user cancels or Meta returns an error, save a recoverable failure state and show a human-readable retry path.

Suggested customer states

Internal state	Customer label	Meaning
NOT_CONNECTED	Connect WhatsApp	No usable connection exists.
CONNECTING	Connecting...	Meta flow started / server provisioning in progress.
ACTION_REQUIRED	Complete setup in Meta	Meta/customer action is still required.
CONNECTED	WhatsApp connected	Provisioning and connection checks passed.
ERROR_RETRYABLE	Try again	Safe to retry without support.
SUPPORT_REQUIRED	Get help	Blocked by account/Meta condition that cannot be automatically resolved.
DISCONNECTED	Reconnect	Previously connected account is no longer usable.

Automatic post-signup provisioning

This is the core differentiator. Once Embedded Signup completes, Slotly should automatically perform the supported server-side tasks required to make the customer's connection usable, following Meta's current Tech Provider onboarding documentation.

Provisioning action	Engineering rule
Validate output	Confirm required business/WABA/phone identifiers are present and tied to the correct onboarding session.
Persist connection	Create or update one canonical WhatsApp connection record for the tenant.
Credential handling	Exchange/store the appropriate server-side credential material using Meta's supported flow. Encrypt sensitive values at rest and never return them to the browser.
Subscribe / configure	Perform the supported API steps required for the customer's WABA/phone to deliver relevant webhook events to Slotly.
Health check	Confirm Slotly can query the expected connection/phone state and that the account is not obviously incomplete.
Idempotency	If the callback or provisioning job runs twice, it must converge on one valid connection rather than duplicate records/subscriptions.
Final state	Mark CONNECTED only after the required technical checks pass.

Important implementation rule

Treat onboarding as a state machine, not one long HTTP request. Meta/API calls can fail or require retry. Persist progress after each significant step and use a background worker for retryable provisioning actions.

Webhooks, tenant routing and send/receive proof

1. Receive Meta webhook requests at one stable production endpoint.
2. Validate request authenticity/signature according to Meta requirements before processing business data.
3. Persist an event fingerprint / Meta message-event ID where available to prevent duplicate processing.
4. Resolve the event's phone number / WABA identifiers to exactly one Slotly tenant.
5. Log the event against that tenant without exposing one tenant's data to another.
6. For the MVP test, reply with a deterministic message such as "Connection successful".
7. Persist outbound send request/result and message status events for debugging.
8. Acknowledge inbound webhooks quickly; perform slow processing asynchronously.

Minimum proof sequence

Step	Event	Pass condition
1	External customer sends "Hello"	Message reaches Meta / the connected business number.
2	Meta sends webhook	Slotly receives and verifies the event.
3	Router resolves tenant	Correct business is identified using Meta connection identifiers.
4	Slotly sends test reply	Outbound message is accepted by Meta.
5	Delivery/status observed	Slotly records the resulting status event where provided.

External-business validation

Do not call the onboarding layer complete after testing only with SlotlyDev-owned assets, developers or test users. Run it against genuinely separate business assets.

Test	What must be proven
Test A - new/fresh number path	Unrelated business completes signup -> number becomes connected -> inbound webhook reaches Slotly -> Slotly sends reply. Record every manual intervention; target is zero Slotly configuration work.
Test B - existing WhatsApp Business App path	Use an eligible unrelated business that already uses the WhatsApp Business app. Verify the current Meta Business App onboarding path, what the customer sees, whether the app remains usable as expected, and any eligibility restrictions.
Test C - failure/retry	Cancel halfway, use a non-eligible account/number, fail verification, retry, reconnect, and confirm Slotly never creates cross-tenant or duplicate state.

Business App promise

Do not market “keep your existing WhatsApp Business app and number” as universally supported until Test B passes on the production configuration and the current Meta eligibility rules are understood.

Pilot-ready connection dashboard

Once the underlying connection works, build the smallest dashboard that gives a non-technical owner confidence. The dashboard should communicate business outcomes and status, not API concepts.

UI element	What the customer sees
Connection card	Connected / action required / error; business display name; masked WhatsApp number; last successful health check.
Test message	A guided send/receive test so the business can verify the connection without reading logs.
Reconnect	Safe reauthorization/reconnection path if Meta access is revoked or the connection becomes invalid.
Setup progress	Simple progress indicator: Account -> WhatsApp -> Test -> Ready.
Help	Contextual error-specific help. Generic "contact support" should be the final fallback, not the first response.

Never show by default

- WABA ID
- Phone Number ID
- access tokens
- webhook URLs
- Graph API versions
- raw Meta error JSON
- permission scopes

First sellable service after onboarding works

Add a rule-based “Business Info + Menu” automation before adding AI. It is cheap to operate, predictable, easy to test and solves repetitive questions for service businesses.

Automation capability	MVP behaviour
Welcome	Custom greeting and menu when a customer initiates a conversation.
Services & prices	Structured list entered by the business owner.
Booking / enquiry link	Fresha, website, form or other booking link.
Hours	Trading hours and closed/open information.
Location	Address / map link.
Lead capture	Name + requested service + optional contact/notes.
Human handover	Stop automation and flag/notify the business when a person is required.

Do not lead with AI

AI should be a later paid capability with usage controls. The first commercial version should prove that the connection layer and simple automation deliver value before adding inference cost and prompt/knowledge-base complexity.

Minimum data model

Entity	Key fields	Purpose
businesses	tenant id, name, status, owner	Root boundary for tenant isolation.
users	identity, business membership, role	Access control.
whatsapp_connections	tenant, Meta business/WABA/phone IDs, status, credential reference	Canonical connection record.
onboarding_sessions	tenant, state/nonce, step, timestamps, error category	Recoverable state machine.
webhook_events	tenant, event fingerprint, type, timestamps, processing state	Deduplication and debugging.
outbound_messages	tenant, request id, destination, status, Meta id	Send tracking and future billing/usage.
audit_log	actor, tenant, action, timestamp, safe metadata	Security and support traceability.

Security controls required before pilot

- Strict tenant scoping on every database query and background job.
- Credentials/tokens encrypted at rest; never rendered to client-side JavaScript or logs.
- Webhook authenticity/signature validation and replay/deduplication controls.
- CSRF/state/nonce protection around onboarding flow and callback/event handling.
- Least-privilege admin/support access with audit records for sensitive actions.
- PII minimization: store only what the product needs.
- Central secret/config management; no secrets committed to source control.
- Redacted structured logs: IDs may be logged where operationally necessary, credentials may not.
- Backups and tested restore procedure for tenant and connection state.

The onboarding must fail safely

Scenario	Expected system state	Required behaviour
User closes Meta flow	Session remains incomplete; no Connected state.	Offer Resume/Restart without duplicates.
Wrong Meta permissions/access	Meta rejects or returns insufficient access.	Explain which account/admin action is required.
Phone verification fails	No usable phone connection.	Keep ACTION_REQUIRED; do not provision partially as Connected.
Number already linked/unsupported state	Meta reports conflict/eligibility issue.	Classify as support/action required; preserve diagnostic code safely.
Provision API call times out	Unknown result.	Retry idempotently in background; never duplicate subscriptions/connection records.
Webhook delivered twice	Duplicate event.	Deduplicate and process once.
Webhook signature invalid	Untrusted request.	Reject and log security event.
Tenant mapping missing	Event cannot be resolved.	Quarantine for investigation; never guess tenant.
Credential revoked/expired	Previously connected tenant stops working.	Mark degraded/disconnected and present reconnect path.
Meta account restricted	Messaging/provisioning blocked.	Surface account-level action required; do not promise automatic bypass.

Pilot with 5 real businesses

Use a deliberately small pilot to observe where ordinary business owners struggle. The point is not revenue maximization; it is to measure whether the onboarding really behaves like SaaS.

Pilot rule	Detail
Pilot mix	At least 5 unrelated businesses. Prefer service businesses that already use WhatsApp heavily; include at least one existing WhatsApp Business App user if eligible.
Observation rule	Every time the customer contacts Slotly for help, record the exact step, Meta screen/error and whether product guidance could prevent it.
No hidden manual fixes	If the operator changes Meta settings behind the customer's back, mark that onboarding as assisted/failed for the self-service metric.

Metrics that matter first

Metric	Initial target	Why it matters
Self-service completion rate	Target $\geq 80\%$ in early pilot, then improve	Percentage that reach Connected without Slotly manual Meta configuration.
Median time to connected	Target < 10 min where Meta/account prerequisites are ready	Measures the core product promise.
Provisioning success rate	Target $> 95\%$ after Meta flow completes	Separates Slotly engineering failures from Meta/customer eligibility failures.
Support contacts per onboarding	Drive toward < 0.2	Critical to low-cost SaaS economics.
Webhook/send-receive verification	100% of Connected tenants	Never mark Connected if messaging proof is absent.
Retry recovery rate	Track by failure category	Shows whether failures are recoverable without staff.

Recommended build sequence and estimate

Planning estimate for one developer working consistently with existing web-development tooling: roughly 4-6 weeks to reach a credible pilot-ready onboarding foundation. This is not a fixed delivery promise; Meta account restrictions, Business App onboarding eligibility and production review/configuration issues can extend it.

Timebox	Focus	Exit criteria
Week 1	Meta production readiness + data model + connection states	Gate 0 passed; reproducible app/config setup.
Week 2	Slotly auth/tenant + Embedded Signup session integration	External flow can launch and return/capture valid onboarding output.
Week 3	Post-signup provisioning + secure credential storage + idempotent jobs	A completed signup can be automatically provisioned.
Week 4	Webhook verification/routing + outbound send + external new-number test	Unrelated business sends/receives successfully.
Week 5	Business App onboarding test + failure/retry hardening + connection dashboard	Pilot flow handles common failure modes.
Week 6	5-business pilot + instrumentation + fixes	Decision on whether onboarding is ready to productize.

Backlog after the connection MVP

1. Business profile: services, prices, hours, location, booking link.
2. Rule-based welcome/menu automation.
3. Human handover and notifications.
4. Subscription billing and usage accounting.
5. Basic inbox / conversation view.
6. Analytics.
7. AI assistant with metered usage.
8. Broadcasts/templates and advanced workflows.
9. External integrations and multi-number/team features.

Main risks and how to control them

Risk	Failure mode	Control
Meta dependency	Policies, API versions, onboarding requirements and eligibility can change.	Keep Meta integration isolated, follow official docs, version API calls intentionally, add operational alerts and avoid hard-coded assumptions.
Onboarding support burden	Low-tech customers may get stuck on Meta screens.	Instrument every step, build specific guidance, use assisted setup only as exception and measure its rate.
Business App onboarding variability	Existing-number/App path may have eligibility or account restrictions.	Test before marketing it as universal; keep a clear supported fallback path.
Credential/security exposure	One mistake could compromise multiple tenants.	Server-side secrets, encryption, tenant isolation, audit logs, least privilege and security review before scale.
Low-price economics	Support + Meta usage + future AI can exceed subscription margin.	Do not finalize pricing until pilot support effort and usage cost are measured.
Feature creep	Trying to match WATI/BizAI immediately delays the differentiator.	Connection -> simple automation -> pilot -> only then expand.

Decisions still to verify before public launch

- Exact current production Embedded Signup version/configuration to use.
- Whether existing WhatsApp Business App onboarding is enabled/eligible for the target South African customer cases.
- Precise Meta billing/customer payment method experience for the chosen onboarding model.
- Reconnection/offboarding lifecycle and how customer data is handled when access is revoked.
- Production rate limits and operational monitoring thresholds.
- POPIA/privacy terms, data retention and customer-facing consent/processor documentation.
- Subscription price only after pilot support and message/AI cost data exists.

Go / no-go rule

Proceed to the larger SaaS only if the pilot proves that most eligible businesses can connect without SlotlyDev manually configuring their Meta account. If manual intervention remains the normal path, fix onboarding before building CRM/AI breadth.

Implementation references

These are the primary Meta documents used to anchor the plan. Because Meta changes WhatsApp onboarding flows and API requirements over time, re-check the current version of these documents during implementation rather than treating this PDF as an immutable API specification.

Become a Tech Provider

Meta for Developers - WhatsApp Solution Providers / Tech Providers

<https://developers.facebook.com/documentation/business-messaging/whatsapp/solution-providers/get-started-for-tech-providers>

Embedded Signup overview

Meta for Developers - WhatsApp Embedded Signup

<https://developers.facebook.com/documentation/business-messaging/whatsapp/embedded-signup/overview>

Embedded Signup implementation

Meta for Developers - implementation guidance

<https://developers.facebook.com/documentation/business-messaging/whatsapp/embedded-signup/implementation>

Onboarding customers as a Tech Provider

Meta for Developers - post-Embedded-Signup onboarding steps

<https://developers.facebook.com/documentation/business-messaging/whatsapp/embedded-signup/onboarding-customers-as-a-tech-provider>

Final recommendation

Start now with the connection MVP. Treat the first external business successfully connected without manual SlotlyDev Meta configuration as the first major product milestone. Do not spend meaningful time on AI, CRM, campaigns or advanced automation until that milestone has been repeated reliably across unrelated businesses.

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